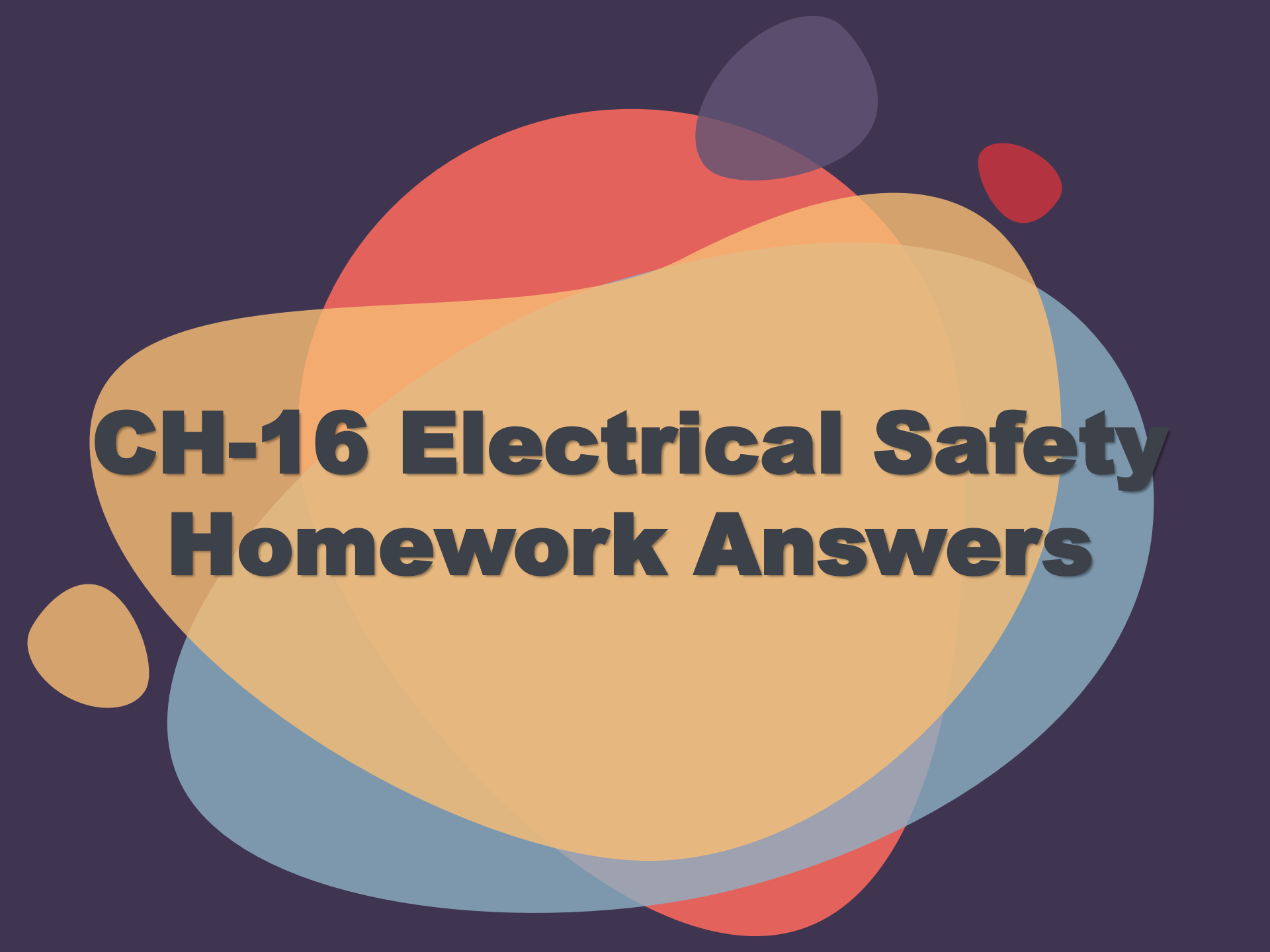




CH-16 Electrical Safety Homework Answers



01

Electrical shock intensity and effects will depend on the:

- A. Age of the conductor
- B. Length of the conductor
- C. Current path, frequency and duration**
- D. Voltage drop across the conductor



02

What is the annual electricity cost to operate a 100 W light bulb all day for every day of the year? Electricity in this jurisdiction costs 10.2 cents per kWh.

A. \$10.20

B. \$19.96

C. \$24.22

D. \$89.35

Consumption = $100\text{W} \times 365\text{day} \times 24\text{h} = 876000\text{Wh} = 876\text{kWh}$

Cost = $876\text{kWh} \times 10.2\text{cent} = 8935\text{cent} = 89.35\text{\$}$



03

Accidental contact with 60 hertz current of 5 to 25 milliamperes can cause which of the following?

- A. Death in most adults
- B. Ventricular fibrillation
- C. Paralysis**
- D. Harmless exposure



04

How does a ground fault circuit interrupter work?

A. Senses small amount of current

B. Senses small amount of heat

C. Senses dampness

D. Senses bad times



05

As a safety professional you are notified that some rigid metal conduit is warm to the touch. What is your suspicion?

- A. Short in system
- B. Transformer is undersized
- C. Aluminium wire
- D. Current flow is too high**



06

What is the primary purpose of NFPA 70E standards?

- A. To provide guidelines for petrochemical plant operations
- B. To detail the financial responsibilities of electrical contractors
- C. To outline safety procedures for working with electrical systems and equipment**
- D. To establish the color codes for wiring in residential buildings



07

There are 3 resistors wired together in parallel.

$R_1 = 500 \, \Omega$, $R_2 = 1000 \, \Omega$, and $R_3 = 1500 \, \Omega$.

What is the total resistance of this circuit?

A. $0.0037 \, \Omega$

B. $272 \, \Omega$

C. $1,000 \, \Omega$

D. $3,000 \, \Omega$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{500} + \frac{1}{1000} + \frac{1}{1500} = 0.002 + 0.001 + 0.00067 = 0.00367$$

$$R_{\text{parallel}} = \frac{1}{0.00367} = 272$$



08

How many kilowatts of power are delivered to a 400 V welding machine that supplies 70A to the circuit?

A. **28kW**

B. 5.72 kW

C. 2,800 kw

D. 28,000 kw

$$P = VI = 400V \times 70A = 28000W$$

$$P = 28kW$$



09

Which statement best describes the principle of operation employed by a GFCI?

- A. The GFCI detects minute fluctuations in the conductivity of the neutral wire in an AC circuit, triggering a fast-acting switch that interrupts the supply of electricity.
- B. The GFCI detects changes in resistance of the supply wire in an AC circuit, triggering a fast-acting switch that interrupts the supply of electricity.
- C. The GFCI detects line voltage differences between the supply wire and the ground wire in an AC circuit, triggering a fast-acting switch that interrupts the supply of electricity.
- D. The GFCI detects line amperage differences between the supply wire and the ground wire in an AC circuit, triggering a fast-acting switch that interrupts the supply of electricity.**



10

Overloading an electrical circuit is a known fire hazard. This hazard exists due to the heat produced by the overload situation that can be a source of ignition. Overloading a circuit is a greater hazard because:

- A. **The heat breaks down the electrical insulation which can lead to short circuits**
- B. The device connected to the circuit can fail due to a brown-out condition
- C. The circuit breaker can fail
- D. The heat from the circuit can ignite the insulation of the wire